Vaibhav Hiwase



AI/ML Engineer & Technical Lead — LLM Applications, Evaluation & Production AI Platforms

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PROFESSIONAL SUMMARY

AI/ML engineer and technical lead with 8+ years building LLM applications and data platforms that run in production. Currently own two production AI systems on Palantir Foundry & AIP: an employee benefits assistant built on Agent Studio, AIP Logic and Foundry Functions, and an automated document-verification workflow on Foundry Automate, along with the AIP Evals suites, guardrails, observability and Marketplace release tooling behind them. Earlier work covers contract-document AI with OpenAI, Claude and Llama models, distributed service architecture on AWS Step Functions and Redis, Databricks pipelines, and Kubernetes deployments on Azure and AWS.

TECHNICAL SKILLS

- **Palantir AIP:** AIP Agent Studio, AIP Logic, AIP Evals, Automate, Foundry ML (models, inference), AI FDE (AI-assisted development)
- **Foundry Ontology & Apps:** Ontology modeling (object, link and action types, interfaces, function-backed actions), Functions (Python/TypeScript), OSDK & Developer Console, Compass API, Workshop, Slate, Contour, Fusion Sheets
- **Foundry Data & DevOps:** Pipeline Builder, Code Repositories & Workspaces (Python transforms), Data Connection (streaming, CDC), Media Sets, Restricted Views & Markings, Global Branching, Marketplace & DevOps, multi-environment releases
- **AI, LLM & Gen-AI:** OpenAI, Claude, Gemini, Llama, VLLM, LangChain, LlamaIndex, RAG, embeddings, vector search (FAISS, Pinecone, Weaviate, Chroma), agentic workflows, Model Context Protocol (MCP), LLM A/B testing, reinforcement learning
- **LLM Application Engineering:** conversational agents, RAG, intent classification and routing, entity extraction, tool/function calling, guardrails, response and citation validation, hallucination mitigation, prompt optimisation, streaming responses, token and cost tracking
- **LLM Evaluation:** LLM-as-a-Judge design and judge–expert alignment, critique shadowing, binary pass/fail rubrics, eval dataset curation, error analysis, regression harnesses, release gating
- **Architecture & Systems:** microservice decomposition, event-driven and asynchronous processing, message brokers and queues (Redis, AWS SQS, RQ), workflow orchestration (AWS Step Functions, Foundry Automate), parallel and chunked processing, autoscaling (KEDA), API gateway and auth (Azure APIM, OAuth 2.0 / OpenID Connect)
- **Computer Vision & Document AI:** Azure Form Recognizer, Azure Custom Vision, Mask R-CNN, layout and table detection, document parsing, image classification, blur and tamper detection
- **Data Engineering:** Spark, PySpark, Polars, DuckDB, Delta Lake, MLflow, Databricks, Unity Catalog, DBFS, incremental pipelines, schema evolution, scalable ETL design, governance and lineage, cluster management
- **Programming:** Python (Advanced), PySpark, Polars, Pandas, NumPy, PyTorch, Scikit-learn, Pydantic, TypeScript, SQL
- **Cloud & MLOps:** Azure (AKS, ML Studio, Key Vault, APIM, Container Apps), AWS (SQS, Step Functions, S3), Kubernetes, Docker, TorchServe, Ollama, GitHub Actions
- **Databases:** MySQL, PostgreSQL, Redis (caching, queues, message broker), vector DBs (FAISS, Pinecone, Chroma, Weaviate)
- **Observability & Monitoring:** Arize Phoenix (LLM & pipeline observability), RQ Dashboard, Opik by Comet
- **Tools:** Git, Docker, Jupyter Notebook, VS Code, Cursor, Antigravity

EXPERIENCE

Lead AI/ML Engineer — Palantir Foundry & AIP

Innovecture (Client: Aptia Group) · Remote, India — Dec 2025 – Present

Build an AI assistant that answers employees' benefits questions, automated checks for the documents they submit and a framework that tests the AI's answers. Serve as Platform Admin, providing L3 (top-level) production support across 5 environments.

GenAI Benefits Assistant

- Raised coverage-answer accuracy from **68.9% to 97.9%** by replacing about 30 rigid lookup tables with a principle-based prompt on Gemini Flash, which also cut token cost 28% and worst-case response time 48%.
- Reached **99.4%** routing accuracy by rebuilding the AI classifier that sorts each question into one of 26 topics.
- Reduced made-up answers (hallucinations) in the assistant's document-based (RAG) replies by having it cite its sources and run an 11-point self-check before answering.

AI Evaluation & Testing

- Designed a **532-question** test suite from real plan documents, including 388 questions written by business users, with an AI judge that must agree with human experts at least 90% of the time.
- Built an AI document search (embedding-based RAG) that matches each test question to the right person and pages, and cut its workload **99.9%** by removing duplicate pages.

- Found **85 of 182** business test questions that the source documents could not answer, using a keyword matcher, and removed a faulty matching step that produced 60 false matches.
- Added analytics for accuracy, cost and speed on every test run, and a Helpfulness score that broke no existing tests.

Document Verification Automation

- Stopped verification requests from getting stuck when a person's data was not yet available, with a readiness check across **22 automations** that retries every 15 minutes and escalates to manual review after 24 hours.
- Replaced silent failures with a clear, auditable result for unsupported documents and clients excluded from AI review.
- Made **24 of 36** untested document scenarios testable with a clone action that re-runs checks without re-uploading.
- Built end-to-end load tests that measure response times (**p50–p99**), throughput and error rates at up to 250 times baseline load, from Foundry actions to AWS SQS delivery.

Data Engineering & Pipelines

- Replaced a failing Postgres data feed with an incremental Spark pipeline that matched the source exactly on **8M+ rows**.
- Fixed recurring crashes in the Pendo analytics feed, which now finishes in **14.5 minutes**, by processing one day at a time and adapting to new columns automatically.
- Corrected dependent-coverage counts that were **nearly three times** too high by switching to Foundry-calculated data.
- Repaired and expanded the assistant's monitoring from 16 to 21 log sources, storing user messages only in hashed form.

Platform Engineering & DevOps

- Built a deployment-checking tool, backed by **158 unit tests**, that compares 36 types of Foundry resources across 5 environments and highlights differences in an interactive dashboard.
- Made Foundry Marketplace packages install cleanly in every environment by removing hard-coded IDs and a dependency that broke the Stage environment (verified on **22.7M rows**).
- Set up automatic email alerts for failed builds, data syncs and search indexing, with separate alerts for production.
- Restored **70+ of 107** user actions, such as create, edit and delete, in the development environment.
- Flagged two data-privacy risks and designed test pipelines to store document fingerprints instead of document content.

Technical Lead — ML & AI

TechnoMile · Remote, India — Sept 2023 – Nov 2025

- Built the AI extraction engine behind TechnoMile's **Contract Lifecycle Management** and **Pre-Award** products, pulling clauses, CLINs and sections out of long federal contract PDFs with **OpenAI, Claude and Llama** models. Raised CLIN accuracy from **58% to 90%** across 7 contract domains.
- Shipped contract question-answering with **RAG**, chunking and embedding documents into a **FAISS/Chroma** vector store so answers are grounded in retrieved text rather than model memory.
- Re-architected a monolithic extraction service into four independent CLIN, Clause, Section and Form services orchestrated with **AWS Step Functions**, cutting processing time **40%** through parallel chunk handling and asynchronous PDF-to-image conversion.
- Built the review tooling for QA: a field-by-field accuracy report against ground truth (**Python, Plotly, DataTables**) and a viewer that boxes the exact PDF location each extracted value came from, so reviewers could catch fields the model invented.
- Ran models in production with **MLflow** for experiment tracking and versioning and **Arize Phoenix** to trace prompts, token usage and latency.
- Built Pre-Award Copilot features extracting clauses, deliverables and risk indicators from government solicitations, plus **Model Context Protocol (MCP)** agents that answer questions from internal APIs and databases in plain language.

Senior Consultant

Celebal Technologies · Remote, India — Sept 2021 – Aug 2023

Data science and ML engineering for enterprise clients: document intelligence, production ML infrastructure, and a forecasting platform built with Microsoft.

- Led development of a payslip forgery detection method reaching **85.23% accuracy**, using **Azure Form Recognizer** for extraction.
- Trained payslip classification and blur detection models on **Azure Custom Vision**, reaching **90.87%** and **95.72%** accuracy.
- Built key information extraction and document similarity capabilities across client engagements, and a Hinglish language model for a client use case.
- Architected a secure, high-throughput serving platform on **Redis, Kubernetes/AKS, Docker, Azure API Management, Key Vault and OAuth 2.0**, cutting processing time **10x**.
- Moved long-running document work onto background workers behind a **Redis** queue, with **KEDA** autoscaling on Azure Kubernetes Service and containerised deployment via **Docker**.
- Partnered with **Microsoft** to build a forecasting platform for energy portfolio management, adding training-loss monitoring across **DeepMC, TFT, N-BEATS and TCN** models with pipelines running on **Azure ML Studio** and **Azure Functions**.
- Fine-tuned sentence-embedding models on retail catalog data to improve ad and product recommendations from search queries.
- Mentored the team across document intelligence and language modelling, and supervised a computer-vision project counting exercise repetitions from mobile and IoT camera video.

Machine Learning Engineer

Konverge.AI · India — Aug 2018 – Aug 2021

- Led a team of **10** on solution architecture and work breakdown for Smart Content Extraction.
- Built data ingestion APIs with **Airbyte** connectors for the WEAV Core libraries.
- Built a clustering-based method for extracting paragraphs and tables from documents.
- Trained a **Mask R-CNN** model to classify fungus, yeast and bacteria at pixel level on stained nail-tissue images.
- Built the backend for media analytics: keyword identification, clustering and recommendations.
- Built a **GIS** system that maps geographic locations found in scanned government documents.

PROJECTS

Contract Document AI — TechnoMile

- Trained a layout detection model on **Azure Custom Vision** to find CLIN tables in contract PDFs: **94%** accuracy on 50 annotated documents, rising above **98%** after scaling the training set to 264 with automated retraining.
- Built a reusable ML pipeline library with separate training, inference and deployment components, deployable on **TorchServe, Ollama or VLLM**.
- Wrote a verification tool that matches each extracted value back to its place in the document and scores confidence, so unsupported fields get caught.
- Cut CLIN processing from **370s to 220s** with chunk-level parallelism, keeping memory flat by converting PDF pages to images one at a time.

Large Language Model Training

- Fine-tuned a Spanish sentence-embedding model (**Sentence Transformers**) on Walmart Mexico catalog data to improve ad recommendations from search queries.
- Used **FLAN-T5 Large** to generate search keywords from product text, improving those recommendations further.
- Used **FAISS** to make search lookups faster and lighter on memory.
- Ran the work on **Databricks** and validated results against add-to-cart data.

Payslip Forgery Detection — Celebal Technologies

- Led the work end to end for a payroll client: research, architecture, engineering and monitoring.
- Layered metadata and pixel-level checks on top of the forgery detector to catch manipulated documents that passed content extraction.
- Built the **Flask** and **React** application, secured with **OAuth 2.0** and **OpenID Connect** behind **Azure API Management**, shipped through **GitHub Actions** to **Azure Container Apps**, with an **RQ dashboard** for queue visibility.

EDUCATION

Master of Technology, Computer Science and Engineering — RCOEM, Nagpur, India — Aug 2016 – Aug 2018

Bachelor of Engineering, Information Technology — SVPCET, Nagpur, India — Aug 2012 – Aug 2016

CERTIFICATIONS

STATSX0001: Statistical Learning, StanfordOnline · OAuth 2.0 · Docker · Data Science

ACHIEVEMENTS & ACTIVITIES

Microsoft AI/ML competition (2022) · Artificial Intelligence and Machine Learning Workathon (2021) · Organized and led CSI events (2013–2016) · National Cadet Corps (NCC) cadet with 'A' certificate (2008–2009)